Table 1: Quantitative comparisons with state-of-the-art (SOTA) methods on five public polyps datasets are presented. We mark the best results with **bold** and the second-best scores with underline.

Methods	Published Venue	CVC-ClinicDB		Kvasir		CVC-ColonDB		ETIS		EndoScene	
		Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU
UNet[18]	MICCAI'15	0.823	0.755	0.818	0.746	0.504	0.436	0.398	0.335	0.710	0.627
PraNet[6]	MICCAI'19	0.899	0.849	0.898	0.840	0.709	0.640	0.628	0.567	0.871	0.797
SANet[24]	MICCAI'21	0.916	0.859	0.904	0.847	0.752	0.669	0.750	0.654	0.888	0.815
MSNet[33]	MICCAI'21	0.915	0.866	0.902	0.847	0.747	0.668	0.720	0.650	0.862	0.796
UACANet[11]	MM'21	0.916	0.870	0.905	0.852	0.783	0.704	0.694	0.615	0.902	0.837
LDNet[31]	MICCAI'22	0.932	0.872	0.912	0.855	0.794	0.715	0.778	0.707	0.893	0.826
SSFormer[22]	MICCAI'22	0.906	0.855	0.917	**0.864**	0.802	0.721	0.796	0.720	0.895	0.827
DCRNet[27]	ISBI'22	0.869	0.800	0.846	0.772	0.661	0.576	0.509	0.432	0.753	0.670
Polyp-PVT[5]	AIR'23	0.937	0.889	0.917	**0.864**	**0.808**	**0.727**	0.787	0.706	0.900	0.833
CFANet[35]	PR'23	0.933	0.883	0.915	0.861	0.743	0.665	0.732	0.655	0.893	0.827
SAM-H[12]	ICCV'23	0.547	0.500	0.778	0.707	0.441	0.396	0.517	0.477	0.651	0.606
SAM-L[12]	ICCV'23	0.579	0.526	0.782	0.710	0.468	0.422	0.551	0.507	0.726	0.676
SAM-Adapter[3]	ICCV'23	0.774	0.673	0.847	0.763	0.671	0.568	0.590	0.476	0.815	0.725
AutoSAM[9]	ArXiv'23	0.751	0.642	0.784	0.675	0.535	0.418	0.402	0.308	0.829	0.739
SAMPath[29]	MICCAIw'23	0.750	0.644	0.828	0.730	0.632	0.516	0.555	0.442	0.844	0.756
SAMed[30]	ArXiv'23	0.404	0.273	0.459	0.300	0.199	0.115	0.212	0.126	0.332	0.202
SAMUS[14]	ArXiv'23	0.900	0.821	0.859	0.763	0.731	0.597	0.750	0.618	0.859	0.760
SurgicalSAM[28]	AAAI'24	0.644	0.505	0.740	0.597	0.460	0.330	0.342	0.238	0.623	0.472
MedSAM[15]	Nature'24	0.867	0.803	0.862	0.795	0.734	0.651	0.687	0.604	0.870	0.798
Ours	Efficient-SAM[25]	0.942	0.891	0.914	0.849	0.782	0.680	0.854	0.758	0.900	0.819
Ours	ViT-B	0.950	0.905	0.914	0.848	0.792	0.694	0.856	0.764	0.914	0.843
Ours	ViT-H	**0.951**	**0.906**	**0.920**	0.858	0.799	0.701	**0.861**	**0.769**	**0.919**	**0.852**

not particularly noticeable, the improvement on CVC-ClinicDB, ETIS, and EndoScene is quite significant. Especially, Polyp-PVT[5] showed good performance on all datasets with average Dice and IoU of 0.870 and 0.804, respectively, while our method achieved 0.890 and 0.817, which proves the effectiveness of our model.

Compared to SAM-based (ViT-B) methods, our approach outperformed all others across all datasets. It's important to note that methods like MedSAM[15] and SAMUS[14] still incorporated prompts like SAM, but we removed prompts in the comparative experiment. Besides, despite our efforts, we didn't attain satisfactory results with SurgicalSAM[28] and SAMed[30], potentially due to incompatible training hyperparameters. Moreover, we observed that SAMUS[14] also utilized a CNN auxiliary branch and achieved excellent results, respectively, further validating the effectiveness of the CNN branch. Additionally, with the development of the lightweight model of SAM, we successfully combined our method with EfficientSAM[25] and achieved satisfactory results.

In Fig. 3, we use Fourier analysis as a toolkit to show the difference between features from two encoders. The Fourier spectrum and relative log amplitudes of the Fourier transformed feature maps indicate that the CNN branch captures more high-frequency signals than the ViT baseline. We also provide the qualitative results in Fig. 4, where our predictions are closer to the ground truth.

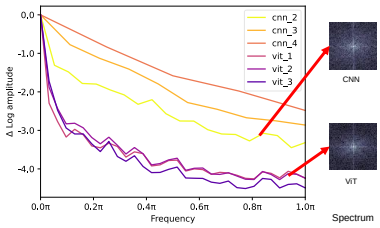

Fig. 3: Relative log amplitudes.

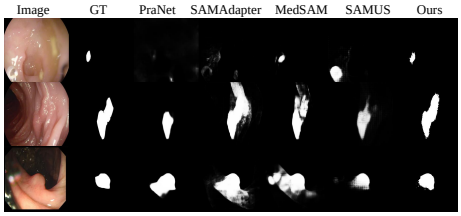

Fig. 4: Several qualitative results.

Table 2: Ablation experiments on five public polyp datasets.

CFA	UPR	CVC-ClinicDB Dice	CVC-ClinicDB IoU	Kvasir Dice	Kvasir IoU	CVC-ColonDB Dice	CVC-ColonDB IoU	ETIS Dice	ETIS IoU	EndoScene Dice	EndoScene IoU
		0.537	0.388	0.652	0.469	0.397	0.268	0.387	0.265	0.525	0.368
✓		0.913	0.843	0.887	0.819	0.775	0.684	0.811	0.708	0.901	0.831
	✓	0.558	0.407	0.672	0.519	0.415	0.282	0.382	0.360	0.610	0.453
✓	✓	0.950	0.905	0.914	0.848	0.792	0.694	0.856	0.764	0.914	0.843

Table 3: Ablation studies on EPA.

CFA			UPR	CVC-ClinicDB Dice	CVC-ClinicDB IoU	Kvasir Dice	Kvasir IoU
CA	Fusion	PE					
			✓	0.558	0.407	0.672	0.519
✓			✓	0.838	0.737	0.866	0.772
✓	✓		✓	0.941	0.891	0.911	0.848
✓	✓	✓	✓	0.950	0.905	0.914	0.848

Table 4: Ablation studies on UPR.

CFA	UPR			CVC-ClinicDB Dice	CVC-ClinicDB IoU	Kvasir Dice	Kvasir IoU
	TN	NN	Hint				
✓				0.913	0.843	0.887	0.819
✓	✓			0.936	0.885	0.912	0.859
✓		✓		0.944	0.896	0.914	0.850
✓	✓	✓		0.936	0.881	0.875	0.814
✓		✓	✓	0.950	0.905	0.914	0.848

Ablation Study. We conducted ablation experiments to verify the effectiveness of the proposed CFA and UPR. For our baseline, we use ViT-B as the backbone of SAM and remove the prompt encoder. As shown in Table 2, we are confident to assert that the contribution of each module to the overall performance enhancement is significant, and their combination produces the best overall performance.

To ensure the reliability of the CFA, we conducted ablation experiments on the introduced cross-branch attention (CA), multi-level features fusion (Fusion), and position embedding replacement (PE). As shown in Table 3, compared with the baseline, our method achieved better performance and improved the mean Dice score by 31.7% and the mean IoU score by 41.4%. In addition, we tried to train (1) the Transformer block norm (TN), (2) the neck layer norm (NN), and (3) both to validate the effectiveness of UPR. The results are shown in Table 4. Due to the limitation of computing power, we set the batch size for experiments involving TN to 2, while the others were set to 4. Currently, only training the neck layer norm demonstrated superior performance.

4 Conclusion

We introduce a novel method called ASPS for polyp segmentation, designed to address the limitations of the SAM model in capturing information and bridging the domain gap between endoscopy images. The CFA module incorporates a trainable CNN encoder branch to supplement the frozen ViT encoder, integrating multi-scale and multi-level features. Additionally, the UPR module reduces uncertainty during training by introducing hints and adjusting the normalization layer, promoting the adaptation of the model in the endoscopy field. Through experiments on five common polyp datasets, we verify the effectiveness and superiority of our method. To extend our work, our future direction focuses on investigating more efficient methods using SAM, enabling us to fully harness the foundation model for effective polyp segmentation.

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Disclosure of Interests. We declare that we have no competing interests.

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